

Ayudantía 13- MAT1620

1. Calcule las siguientes integrales iteradas

(a) $\int_1^4 \int_0^2 (6x^2y - 2x) dy dx.$

(b) $\int_{-3}^3 \int_0^{\pi/2} (y + y^2 \cos(x)) dx dy.$

(c) $\int_0^1 \int_0^1 xy \sqrt{x^2 + y^2} dy dx.$

2. Evalúe las siguientes integrales.

(a) $\iint_D x \cos(y) dA$, donde D está acotada por $y = 0$, $y = x^2$ y $x = 1$.

(b) $\iint_D y^2 dA$, donde D es la región triangular con vértices $(0, 1)$, $(1, 2)$ y $(4, 1)$.

(c) $\iint_D xy^2 dA$, donde D está acotada por $x = 0$ y $x = \sqrt{1 - y^2}$.

3. Determine el volumen del sólido encerrado por los cilindros $z = x^2$, $y = x^2$ y los planos $z = 0$, $y = 4$.

4. Invierta el orden de integración para calcular el valor de

$$\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} dy dx$$

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$$\begin{aligned} a) \int_1^4 \int_0^2 (6x^2y - 2x) dy dx &= \int_1^4 \left. \frac{6x^2y^2}{2} - 2xy \right|_0^2 dx = \int_1^4 (3x^2y^2 - 2xy) \Big|_0^2 dx \\ &= \int_1^4 (12x^2 - 4x) dx = \left. 4x^3 - 2x^2 \right|_1^4 = 256 - 32 - 4 + 2 = 222 \end{aligned}$$

$$\begin{aligned} b) \int_{-3}^3 \int_0^{\pi/2} (y + y^2 \cos(x)) dx dy &= \int_{-3}^3 \left. xy + y^2 \sin(x) \right|_0^{\pi/2} dy \\ &= \int_{-3}^3 \left(\frac{\pi}{2} y + y^2 \right) dy = \left. \frac{\pi}{4} y^2 + \frac{1}{3} y^3 \right|_{-3}^3 = \frac{9\pi}{4} + 9 - \frac{9\pi}{4} + 9 = 18 \end{aligned}$$

$$c) \int_0^1 \int_0^1 xy \sqrt{x^2 + y^2} dy dx \quad \begin{matrix} \curvearrowright u = x^2 + y^2 \\ \Rightarrow du = 2y dy \Rightarrow y dy = \frac{1}{2} du \end{matrix}$$

$$= \int_0^1 \int_{x^2}^{x^2+1} \frac{1}{2} x \sqrt{u} du dx = \int_0^1 \frac{1}{2} x \cdot \frac{2}{3} \cdot u^{3/2} \Big|_{x^2}^{x^2+1} dx$$

$$= \int_0^1 \frac{1}{3} x \left[(x^2+1)^{3/2} - x^3 \right] dx$$

$$= \frac{1}{3} \left[\int_0^1 x (x^2+1)^{3/2} dx - \int_0^1 x^4 dx \right]$$

$$\begin{matrix} \curvearrowright \\ u = x^2 + 1 \\ du = 2x dx \\ \Rightarrow x dx = \frac{1}{2} du \end{matrix}$$

$$= \frac{1}{3} \left[\int_{\frac{1}{2}}^{\frac{5}{2}} \frac{1}{2} u^{3/2} du - \frac{1}{5} x^5 \Big|_0^1 \right]$$

$$\frac{1}{3} \left[\int_1^2 \frac{1}{2} u^{3/2} du - \frac{1}{5} x^5 \Big|_0^1 \right] = \frac{1}{3} \left[\frac{1}{2} \cdot \frac{2}{5} u^{5/2} \Big|_1^2 - \frac{1}{5} \right]$$

$$= \frac{1}{3} \cdot \left(\frac{1}{5} 2^{5/2} - \frac{1}{5} - \frac{1}{5} \right) = \frac{1}{15} \cdot \sqrt{32} - \frac{2}{15}$$

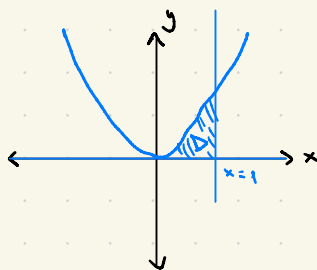
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a) $\iint_D x \cos(y) dA$



$$x \in [0, 1]$$

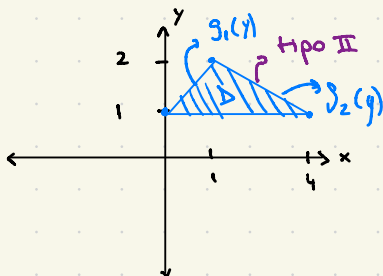
$$0 \leq y \leq x^2$$

$$\int_0^1 \int_0^{x^2} x \cos(y) dy dx = \int_0^1 x \sin(y) \Big|_0^{x^2} dx = \int_0^1 x \sin(x^2) dx$$

$$\underbrace{u = x^2}_{du = 2x dx}$$

$$= \int_0^1 \frac{1}{2} \sin(u) du = -\frac{1}{2} \cos(u) \Big|_0^1 = \frac{1}{2} - \frac{1}{2} \cos(1)$$

b) $\iint_D y^2 dA$



$$1 \leq y \leq 2$$

$$g_1(y) = x = \frac{1-0}{2-1} y + -1$$

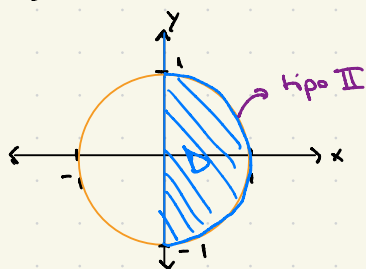
$$g_1(y) = y - 1$$

$$g_2(y) = \frac{4-1}{1-2} y + 7$$

$$\hookrightarrow g_2(y) = -3y + 7$$

$$\begin{aligned}
 \int_1^2 \int_{y-1}^{-3y+7} y^2 dx dy &= \int_1^2 x y^2 \Big|_{y-1}^{-3y+7} dy = \int_1^2 y^2 (-3y+7) - y^2 (y-1) dy \\
 &= \int_1^2 -3y^3 + 7y^2 - y^3 + y^2 dy = \int_1^2 -4y^3 + 8y^2 dy \\
 &= -y^4 + \frac{8}{3} y^3 \Big|_1^2 = -16 + \frac{64}{3} + 1 - \frac{8}{3} = \frac{56}{3} - 15 = \frac{11}{3}
 \end{aligned}$$

c) $\iint_D x y^2 dA$, $D: x=0, x=\sqrt{1-y^2}$



$$\hookrightarrow x^2 = 1 - y^2$$

$$x^2 + y^2 = 1$$

$$-1 \leq y \leq 1$$

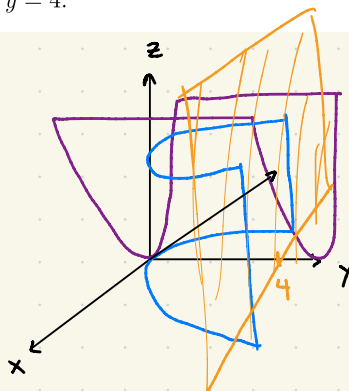
$$0 \leq x \leq \sqrt{1-y^2}$$

$$\int_{-1}^1 \int_0^{\sqrt{1-y^2}} x y^2 dx dy = \int_{-1}^1 \frac{1}{2} x^2 y^2 \Big|_0^{\sqrt{1-y^2}} dy = \int_{-1}^1 \frac{1}{2} (1-y^2) y^2 dy$$

$$= \int_{-1}^1 \frac{1}{2} y^2 - \frac{1}{2} y^4 dy = \frac{1}{6} y^3 - \frac{1}{10} y^5 \Big|_{-1}^1 = \frac{1}{6} - \frac{1}{10} + \frac{1}{6} - \frac{1}{10}$$

$$= \frac{1}{3} - \frac{1}{5} = \frac{2}{15}$$

3. Determine el volumen del sólido encerrado por los cilindros $z = x^2, y = x^2$ y los planos $z = 0, y = 4$.



$$g_1(x, y) = x^2$$

$$g_2(x, y) = 0$$

$$V = \iint_D f(x, y) \, dA$$

$$f(x, y) = g_1(x, y) - g_2(x, y)$$

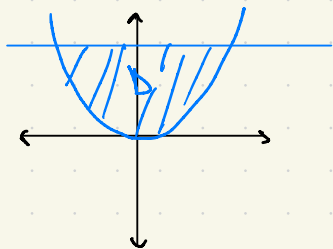
$$\hookrightarrow f(x, y) = x^2 - 0$$

$$\rightarrow \text{si } y = 4 \Rightarrow 4 = x^2$$

$$\hookrightarrow x = \pm 2$$

$$-2 \leq x \leq 2$$

$$x^2 \leq y \leq 4$$



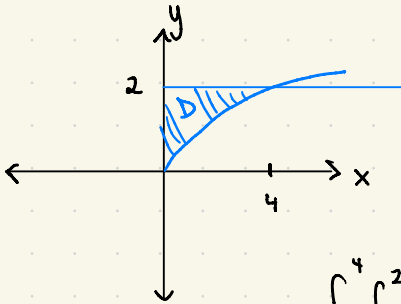
$$V = \int_{-2}^2 \int_{x^2}^4 x^2 \, dy \, dx = \int_{-2}^2 x^2 y \Big|_{x^2}^4 \, dx$$

$$V = \int_{-2}^2 4x^2 - x^4 \, dx = \frac{4}{3} x^3 - \frac{1}{5} x^5 \Big|_{-2}^2$$

$$\Rightarrow V = \frac{32}{3} - \frac{32}{5} + \frac{32}{3} - \frac{32}{5} = \frac{128}{15} \, u^3$$

4. Invierta el orden de integración para calcular el valor de

$$\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} dy dx$$



$$0 \leq y \leq 2$$

$$0 \leq x \leq y^2$$

$$\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} dy dx = \int_0^2 \int_0^{y^2} \frac{1}{y^3 + 1} dx dy$$

$$= \int_0^2 \frac{x}{y^3 + 1} \Big|_0^{y^2} dy = \int_0^2 \frac{y^2}{y^3 + 1} dy \quad \begin{array}{l} \text{blue arrow } u = y^3 + 1 \\ \Rightarrow du = 3y^2 dy \Rightarrow \frac{1}{3} du = y^2 dy \end{array}$$

$$= \int_1^9 \frac{1}{3} \frac{du}{u} = \frac{1}{3} \ln(u) \Big|_1^9 = \frac{2}{3} \ln(3)$$